

Creation Date 05-May-2016

Revision Date 04-Aug-2022

Revision Number 2

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: **Mobile Phase A**  
Cat No. : 227-40102-55, 227-40102-58, 227-40102-91

Unique Formula Identifier (UFI) YTR9-048S-5X0H-1E1Y

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

**Company**

**EU entity/business name**  
Acros Organics BV  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium  
General Info; Tel: +32-14-57 52 11  
(info@acros.com)  
Technical Support; Tel +32-14-56 56 00  
(acros.techsupport@thermofisher.com)

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom  
General info; Tel: +44 (0)1509 231166

**Swiss distributor - Fisher Scientific AG**  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

For customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

**Poison Centre - Emergency information services**

**Ireland : National Poisons Information Centre (NPIC) -**  
**01 809 2166** (8am-10pm, 7 days a week)

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Malta : +356 2395 2000  
Cyprus : +357 2240 5611

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Flammable liquids

Category 3 (H226)

##### Health hazards

Acute oral toxicity

Category 4 (H302)

Acute dermal toxicity

Category 4 (H312)

Acute Inhalation Toxicity - Vapors

Category 4 (H332)

Specific target organ toxicity - (single exposure)

Category 1 (H370)

##### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

### 2.2. Label elements

Contains METHANOL



Signal Word

Danger

#### **Hazard Statements**

H226 - Flammable liquid and vapor

H370 - Causes damage to organs

H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled

#### **Precautionary Statements**

P280 - Wear protective gloves/protective clothing

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P308 + P311 - IF exposed or concerned: Call a POISON CENTER or doctor

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

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## 2.3. Other hazards

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component        | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                            |
|------------------|-----------|-------------------|----------|--------------------------------------------------------------------------------------------------------------|
| Methanol         | 67-56-1   | 200-659-6         | 5-10     | Flam. Liq. 2 (H225)<br>Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>STOT SE 1 (H370) |
| Water            | 7732-18-5 | EEC No. 231-791-2 | 85-90    | -                                                                                                            |
| Ammonium formate | 540-69-2  | EEC No. 208-753-9 | <0.01    | Skin Irrit. 2 (H315)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H335)                                              |

| Component | Specific concentration limits (SCL's)                          | M-Factor | Component notes |
|-----------|----------------------------------------------------------------|----------|-----------------|
| Methanol  | STOT SE 1 (H370) :: C>=10%<br>STOT SE 2 (H371) ::<br>3%<=C<10% | -        | -               |

| Components | Reach Registration Number |
|------------|---------------------------|
| Methanol   | 01-2119433307-44          |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                                                           |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Notes to Physician | Treat symptomatically. |
|--------------------|------------------------|

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## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### **Hazardous Combustion Products**

Nitrogen oxides (NOx).

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Should not be released into the environment.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

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## 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a cool, well-ventilated place. Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

Technical Rules for Hazardous Substances (TRGS) 510  
Storage Class (LGK) (Germany)

Class 3

Switzerland - Storage of hazardous substances

Storage class - SC 3

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component | European Union                                               | The United Kingdom                                                                                              | France                                                                                                                                                                                        | Belgium                                                                                                                                | Spain                                                                                    |
|-----------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Methanol  | TWA: 200 ppm 8 hr<br>TWA: 260 mg/m <sup>3</sup> 8 hr<br>Skin | WEL - TWA: 200 ppm<br>TWA: 266 mg/m <sup>3</sup> TWA<br>WEL - STEL: 250 ppm<br>STEL: 333 mg/m <sup>3</sup> STEL | TWA / VME: 200 ppm (8 heures). restrictive limit<br>TWA / VME: 260 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 1000 ppm.<br>STEL / VLCT: 1300 mg/m <sup>3</sup> .<br>Peau | TWA: 200 ppm 8 uren<br>TWA: 266 mg/m <sup>3</sup> 8 uren<br>STEL: 250 ppm 15 minuten<br>STEL: 333 mg/m <sup>3</sup> 15 minuten<br>Huid | TWA / VLA-ED: 200 ppm (8 horas)<br>TWA / VLA-ED: 266 mg/m <sup>3</sup> (8 horas)<br>Piel |

| Component | Italy                                                                                                               | Germany                                                           | Portugal                                                                                       | The Netherlands                           | Finland                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methanol  | TWA: 200 ppm 8 ore.<br>Time Weighted Average<br>TWA: 260 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>Pelle | 100 ppm TWA MAK;<br>130 mg/m <sup>3</sup> TWA<br>MAKSkin absorber | STEL: 250 ppm 15 minutos<br>TWA: 200 ppm 8 horas<br>TWA: 260 mg/m <sup>3</sup> 8 horas<br>Pele | huid<br>TWA: 133 mg/m <sup>3</sup> 8 uren | TWA: 200 ppm 8 tunteina<br>TWA: 270 mg/m <sup>3</sup> 8 tunteina<br>STEL: 250 ppm 15 minuutteina<br>STEL: 330 mg/m <sup>3</sup> 15 minuutteina<br>Iho |

| Component | Austria                                                                                                                                                       | Denmark                                                           | Switzerland                                                                                                                                       | Poland                                                                            | Norway                                                                                                                                                                          |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methanol  | Haut<br>MAK-KZGW: 800 ppm 15 Minuten<br>MAK-KZGW: 1040 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 200 ppm 8 Stunden<br>MAK-TMW: 260 mg/m <sup>3</sup> 8 Stunden | TWA: 200 ppm 8 timer<br>TWA: 260 mg/m <sup>3</sup> 8 timer<br>Hud | Haut/Peau<br>STEL: 400 ppm 15 Minuten<br>STEL: 520 mg/m <sup>3</sup> 15 Minuten<br>TWA: 200 ppm 8 Stunden<br>TWA: 260 mg/m <sup>3</sup> 8 Stunden | STEL: 300 mg/m <sup>3</sup> 15 minutach<br>TWA: 100 mg/m <sup>3</sup> 8 godzinach | TWA: 100 ppm 8 timer<br>TWA: 130 mg/m <sup>3</sup> 8 timer<br>STEL: 150 ppm 15 minutter. value calculated<br>STEL: 162.5 mg/m <sup>3</sup> 15 minutter. value calculated<br>Hud |

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| Component | Bulgaria                                                      | Croatia                                                                        | Ireland                                                                                                                      | Cyprus                                                                                | Czech Republic                                                                                                  |
|-----------|---------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Methanol  | TWA: 200 ppm<br>TWA: 260.0 mg/m <sup>3</sup><br>Skin notation | kože<br>TWA-GVI: 200 ppm 8 satima.<br>TWA-GVI: 260 mg/m <sup>3</sup> 8 satima. | TWA: 200 ppm 8 hr.<br>TWA: 260 mg/m <sup>3</sup> 8 hr.<br>STEL: 600 ppm 15 min<br>STEL: 780 mg/m <sup>3</sup> 15 min<br>Skin | Skin-potential for cutaneous absorption<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> | TWA: 250 mg/m <sup>3</sup> 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 1000 mg/m <sup>3</sup> |

| Component | Estonia                                                                                                                                                | Gibraltar                                                             | Greece                                                                                                                                  | Hungary                                                                           | Iceland                                                                                                                                             |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Methanol  | Nahk<br>TWA: 200 ppm 8 tundides.<br>TWA: 250 mg/m <sup>3</sup> 8 tundides.<br>STEL: 250 ppm 15 minutites.<br>STEL: 350 mg/m <sup>3</sup> 15 minutites. | Skin notation<br>TWA: 200 ppm 8 hr<br>TWA: 260 mg/m <sup>3</sup> 8 hr | skin - potential for cutaneous absorption<br>STEL: 250 ppm<br>STEL: 325 mg/m <sup>3</sup><br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> | TWA: 260 mg/m <sup>3</sup> 8 órában. AK<br>lehetséges borón keresztül felszívódás | TWA: 200 ppm 8 klukkustundum.<br>TWA: 260 mg/m <sup>3</sup> 8 klukkustundum.<br>Skin notation<br>Ceiling: 400 ppm<br>Ceiling: 520 mg/m <sup>3</sup> |

| Component | Latvia                                                                                | Lithuania                                                   | Luxembourg                                                                                                           | Malta                                                                                            | Romania                                                                 |
|-----------|---------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Methanol  | skin - potential for cutaneous exposure<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> | TWA: 200 ppm IPRD<br>TWA: 260 mg/m <sup>3</sup> IPRD<br>Oda | Possibility of significant uptake through the skin<br>TWA: 200 ppm 8 Stunden<br>TWA: 260 mg/m <sup>3</sup> 8 Stunden | possibility of significant uptake through the skin<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> | Skin notation<br>TWA: 200 ppm 8 ore<br>TWA: 260 mg/m <sup>3</sup> 8 ore |

| Component        | Russia                                                                      | Slovak Republic                                                                  | Slovenia                                                                                                                                | Sweden                                                                                                                                                                       | Turkey                                                           |
|------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Methanol         | TWA: 5 mg/m <sup>3</sup> 1250<br>Skin notation<br>MAC: 15 mg/m <sup>3</sup> | Potential for cutaneous absorption<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> | TWA: 200 ppm 8 urah<br>TWA: 260 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 800 ppm 15 minutah<br>STEL: 1040 mg/m <sup>3</sup> 15 minutah | Indicative STEL: 250 ppm 15 minuter<br>Indicative STEL: 350 mg/m <sup>3</sup> 15 minuter<br>TLV: 200 ppm 8 timmar.<br>NGV<br>TLV: 250 mg/m <sup>3</sup> 8 timmar. NGV<br>Hud | Deri<br>TWA: 200 ppm 8 saat<br>TWA: 260 mg/m <sup>3</sup> 8 saat |
| Ammonium formate | MAC: 10 mg/m <sup>3</sup>                                                   |                                                                                  |                                                                                                                                         |                                                                                                                                                                              |                                                                  |

## Biological limit values

List source(s):

| Component | European Union | United Kingdom | France                                  | Spain                                   | Germany                                                                                                                                           |
|-----------|----------------|----------------|-----------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Methanol  |                |                | Methanol: 15 mg/L urine<br>end of shift | Methanol: 15 mg/L urine<br>end of shift | Methanol: 15 mg/L urine<br>(end of shift )<br>Methanol: 15 mg/L urine<br>(for long-term exposures: at the end of the shift after several shifts ) |

| Component | Italy | Finland | Denmark | Bulgaria | Romania                                |
|-----------|-------|---------|---------|----------|----------------------------------------|
| Methanol  |       |         |         |          | Methanol: 6 mg/L urine<br>end of shift |

| Component | Gibraltar | Latvia | Slovak Republic                                                                                                                     | Luxembourg | Turkey |
|-----------|-----------|--------|-------------------------------------------------------------------------------------------------------------------------------------|------------|--------|
| Methanol  |           |        | Methanol: 30 mg/L urine<br>end of exposure or work shift<br>Methanol: 30 mg/L urine<br>after all work shifts for long-term exposure |            |        |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of

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exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                    | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Methanol<br>67-56-1 ( 5-10 ) |                              | DNEL = 20mg/kg<br>bw/day        |                                | DNEL = 20mg/kg<br>bw/day          |

| Component                    | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Methanol<br>67-56-1 ( 5-10 ) | DNEL = 130mg/m <sup>3</sup>      | DNEL = 130mg/m <sup>3</sup>         | DNEL = 130mg/m <sup>3</sup>        | DNEL = 130mg/m <sup>3</sup>           |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                    | Fresh water     | Fresh water sediment          | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)         |
|------------------------------|-----------------|-------------------------------|--------------------|------------------------------------|----------------------------|
| Methanol<br>67-56-1 ( 5-10 ) | PNEC = 20.8mg/L | PNEC = 77mg/kg<br>sediment dw | PNEC = 1540mg/L    | PNEC = 100mg/L                     | PNEC = 100mg/kg<br>soil dw |

| Component                    | Marine water    | Marine water sediment          | Marine water Intermittent | Food chain | Air |
|------------------------------|-----------------|--------------------------------|---------------------------|------------|-----|
| Methanol<br>67-56-1 ( 5-10 ) | PNEC = 2.08mg/L | PNEC = 7.7mg/kg<br>sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

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|                                        |                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>          | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                  |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particulates filter conforming to EN 143                                                          |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Particle filtering: EN149:2001<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                          |                                                |
|------------------------------------------------|--------------------------|------------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                                |
| <b>Appearance</b>                              | Colorless                |                                                |
| <b>Odor</b>                                    | No information available |                                                |
| <b>Odor Threshold</b>                          | No data available        |                                                |
| <b>Melting Point/Range</b>                     | No data available        |                                                |
| <b>Softening Point</b>                         | No data available        |                                                |
| <b>Boiling Point/Range</b>                     | > 65 °C / 149 °F         |                                                |
| <b>Flammability (liquid)</b>                   | Flammable                | On basis of test data                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                         |
| <b>Explosion Limits</b>                        | No data available        |                                                |
| <b>Flash Point</b>                             | 52 °C / 125.6 °F         | <b>Method -</b> Estimated Literature reference |
| <b>Autoignition Temperature</b>                | No data available        |                                                |
| <b>Decomposition Temperature</b>               | No data available        |                                                |
| <b>pH</b>                                      | No information available |                                                |
| <b>Viscosity</b>                               | No data available        |                                                |
| <b>Water Solubility</b>                        | Miscible                 |                                                |
| <b>Solubility in other solvents</b>            | No information available |                                                |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                                |
| <b>Component</b>                               | <b>log Pow</b>           |                                                |
| Methanol                                       | -0.74                    |                                                |
| <b>Vapor Pressure</b>                          | No data available        |                                                |
| <b>Density / Specific Gravity</b>              | 0.97                     |                                                |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                         |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                                    |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                                |

### 9.2. Other information

|                             |                                        |
|-----------------------------|----------------------------------------|
| <b>Explosive Properties</b> | explosive air/vapour mixtures possible |
|-----------------------------|----------------------------------------|

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available



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## 10.2. Chemical stability

Stable under normal conditions.

## 10.3. Possibility of hazardous reactions

Hazardous Polymerization

No information available.

Hazardous Reactions

None under normal processing.

## 10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition.

## 10.5. Incompatible materials

None known.

## 10.6. Hazardous decomposition products

Nitrogen oxides (NOx).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Category 4

Dermal

Category 4

Inhalation

Category 4

#### Toxicology data for the components

| Component | LD50 Oral                      | LD50 Dermal                   | LC50 Inhalation               |
|-----------|--------------------------------|-------------------------------|-------------------------------|
| Methanol  | LD50 = 1187 – 2769 mg/kg (Rat) | LD50 = 17100 mg/kg ( Rabbit ) | LC50 = 128.2 mg/L ( Rat ) 4 h |
| Water     | LD50 > 90 mL/kg ( Rat )        | -                             | -                             |

##### (b) skin corrosion/irritation;

No data available

##### (c) serious eye damage/irritation;

No data available

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

| Component                    | Test method                                                       | Test species | Study result    |
|------------------------------|-------------------------------------------------------------------|--------------|-----------------|
| Methanol<br>67-56-1 ( 5-10 ) | OECD Test Guideline 406<br>Guinea Pig Maximisation Test<br>(GPMT) | guinea pig   | non-sensitising |

##### (e) germ cell mutagenicity;

No data available

##### (f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

##### (g) reproductive toxicity;

No data available

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| Component                    | Test method             | Test species / Duration          | Study result              |
|------------------------------|-------------------------|----------------------------------|---------------------------|
| Methanol<br>67-56-1 ( 5-10 ) | OECD Test Guideline 416 | Rat / Inhalation<br>2 Generation | NOAEC =<br>1.3 mg/l (air) |

(h) STOT-single exposure; Category 1  
Results / Target organs Optic nerve, Central nervous system (CNS).

(i) STOT-repeated exposure; No data available  
Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity Ecotoxicity effects

| Component | Freshwater Fish                            | Water Flea            | Freshwater Algae |
|-----------|--------------------------------------------|-----------------------|------------------|
| Methanol  | Pimephales promelas: LC50 > 10000 mg/L 96h | EC50 > 10000 mg/L 24h |                  |

| Component | Microtox                                                                        | M-Factor |
|-----------|---------------------------------------------------------------------------------|----------|
| Methanol  | EC50 = 39000 mg/L 25 min<br>EC50 = 40000 mg/L 15 min<br>EC50 = 43000 mg/L 5 min |          |

### 12.2. Persistence and degradability

**Persistence** Persistence is unlikely, based on information available.

| Component                    | Degradability                  |
|------------------------------|--------------------------------|
| Methanol<br>67-56-1 ( 5-10 ) | DT50 ~ 17.2d<br>>94% after 20d |

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Methanol  | -0.74   | <10 dimensionless             |

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

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**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

**Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

**14.1. UN number**

UN1987

**14.2. UN proper shipping name**

ALCOHOLS, N.O.S  
(contains METHANOL)

**Technical Shipping Name**

**14.3. Transport hazard class(es)**

3

**14.4. Packing group**

III

### ADR

**14.1. UN number**

UN1987

**14.2. UN proper shipping name**

ALCOHOLS, N.O.S  
(contains METHANOL)

**Technical Shipping Name**

**14.3. Transport hazard class(es)**

3

**14.4. Packing group**

III

### IATA

**14.1. UN number**

UN1987

**14.2. UN proper shipping name**

ALCOHOLS, N.O.S  
(contains METHANOL)

**Technical Shipping Name**

**14.3. Transport hazard class(es)**

3

**14.4. Packing group**

III

**14.5. Environmental hazards**

No hazards identified

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**14.6. Special precautions for user** No special precautions required

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component        | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|------------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Methanol         | 67-56-1   | 200-659-6 | -      | -   | X     | X    | KE-23193 | X    | X    |
| Water            | 7732-18-5 | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |
| Ammonium formate | 540-69-2  | 208-753-9 | -      | -   | X     | X    | KE-17235 | X    | X    |

| Component        | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|------------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Methanol         | 67-56-1   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Water            | 7732-18-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Ammonium formate | 540-69-2  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

| Component        | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------|-----------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Methanol         | 67-56-1   | -                                                                   | Use restricted. See item 69.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |
| Water            | 7732-18-5 | -                                                                   | -                                                                                                                                        | -                                                                                                     |
| Ammonium formate | 540-69-2  | -                                                                   | -                                                                                                                                        | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

| Component        | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Methanol         | 67-56-1   | 500 tonne                                                                                 | 5000 tonne                                                                               |
| Water            | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           |
| Ammonium formate | 540-69-2  | Not applicable                                                                            | Not applicable                                                                           |

#### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

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Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** Water endangering class = 2 (self classification)

| Component        | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                              |
|------------------|---------------------------------------|------------------------------------------------------|
| Methanol         | WGK 2                                 | Class I : 20 mg/m <sup>3</sup> (Massenkonzentration) |
| Ammonium formate | WGK1                                  |                                                      |

| Component | France - INRS (Tables of occupational diseases)      |
|-----------|------------------------------------------------------|
| Methanol  | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                    | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Methanol<br>67-56-1 ( 5-10 ) | Prohibited and Restricted Substances                                                                           | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed  
H312 - Harmful in contact with skin  
H332 - Harmful if inhaled  
H370 - Causes damage to organs  
H225 - Highly flammable liquid and vapor  
H301 - Toxic if swallowed  
H311 - Toxic in contact with skin  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H331 - Toxic if inhaled  
H335 - May cause respiratory irritation

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

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**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**NOEC** - No Observed Effect Concentration  
**PBT** - Persistent, Bioaccumulative, Toxic

**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (volatile organic compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

|                              |                       |
|------------------------------|-----------------------|
| <b>Physical hazards</b>      | On basis of test data |
| <b>Health Hazards</b>        | Calculation method    |
| <b>Environmental hazards</b> | Calculation method    |

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

|                         |                 |
|-------------------------|-----------------|
| <b>Creation Date</b>    | 05-May-2016     |
| <b>Revision Date</b>    | 04-Aug-2022     |
| <b>Revision Summary</b> | Not applicable. |

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**